Case Study 2:

Hospital Patient Care Analytics Pipeline.

Scenario:

A multi-specialty hospital wants to improve patient care, reduce waiting times, and predict high-risk patients. The hospital collects data from multiple systems such as patient registration, appointment scheduling, laboratory reports, wearable health devices, and doctor consultation notes.
The management wants to build a data-driven solution that can collect, process, store, and analyze patient data to improve operational efficiency and patient outcomes.

Your Task

Using the concepts learned in Data Engineering Fundamentals, design an end-to-end data engineering solution for the hospital.

Data Sources : The hospital collects patient information from multiple systems.

Data Source	**Data Type**
Patient registration	Structured
Blood Bank System	Structured
Appointment scheduling	Structured
LaboratorySystems	Semi-structured
Wearable health devices	Semi-Structured
Billing System	Structured

Data Ingestion

- Batch Processing:Used when data can be collected and processed periodically, such as laboratory reports.
- Stream Processing:Used when data needs to be captured almost immediately, such as wearable readings and appointment updates.
- File-Based Processing:Used for documents, reports, and consultation notes that are uploaded as files.

Datasource	**Ingestion method**	**Reason**
Patient Registration System	Stream Processing	New patient information can arrive continuously
Appointment Scheduling	Stream Processing	Appointment changes should be available
Wearable Health Devices	Stream Processing	Health readings may arrive continuously
Laboratory Systems	Batch Processing	Lab results can be loaded periodically
Medical Documents	File-Based / Batch	collected and loaded periodically are uploaded as files.

ETL Process:
Extract,Transform and Load

Extract

Collect data from:

- Patient Registration System
- Appointment System
- Laboratory System
- Wearable Devices
- Doctor Consultation System
- Medical Documents

Transform

The collected data can be transformed by:

- ❖ Removing duplicate patient records.
- ❖ Validating patient ID
- ❖ Converting wearable measurements into standard units.
- ❖ Validating laboratory results.
- ❖ Categorizing patients based on risk levels.

Load

The transformed data can be loaded into:

- **Data Lake** → raw and large-volume data
- **Data Warehouse** → cleaned, structured, analytics-ready data
- **Database** → operational patient and appointment information

4. Data Storage

Database

Store frequently accessed structured information such as:

- Patient Profiles
- Appointment Information
- Doctor Information
- Department Information
- Basic Laboratory Results

Data Lake

Store raw and large-volume data such as:

- Wearable device data
- Doctor consultation notes
- Medical documents
- Laboratory files
- Patient history

Data Warehouse

Store analytics-ready information such as:

- Patient waiting-time analysis
- Appointment statistics

- Doctor performance reports
- Patient risk analysis
- Laboratory analysis
- Hospital operational reports

5. Data Lake

Zone	Purpose
Raw Zone	Store data as received
Cleansed Zone	Clean and validate data
Curated Zone	Store analytics-ready data
Sandbox Zone	Data science experiments

---

6. Data Warehouse

The Data Warehouse provides a central location for hospital analytics.

Reports Generated

- ❖ Patient Waiting Time Dashboard
- ❖ Total Appointments
- ❖ Laboratory Tests
- ❖ Patient Risk Dashboard
- ❖ Hospital Performance Dashboard

Departments Benefiting

- ❖ Doctors
- ❖ Nursing Department
- ❖ Laboratory Department
- ❖ Hospital Operations
- ❖ Emergency Department
- ❖ Hospital Management

7. Data Modeling and Data Pipeline

A simple data model can be created for the hospital.

Patient Table

- Patient_ID
- Name
- Age
- Gender
- Department

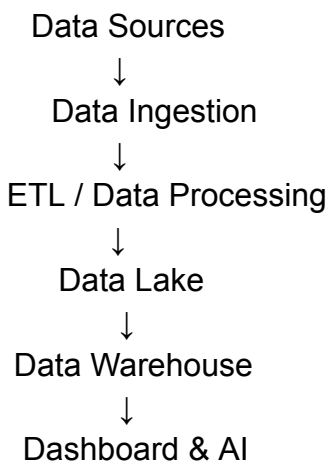
Appointment Table

- Appointment_ID
- Patient_ID
- Appointment_Date
- Appointment_Status
- Waiting_Time

Laboratory Table

- Test_ID
- Patient_ID
- Test_Name
- Test_Result
- Test_Date

End-to-End Pipeline



---

8. Data Quality and Validation

At least five validation rules can be applied.

Validation Rules

- ❖ Patient ID must be unique
- ❖ Appointment date cannot be invalid
- ❖ Laboratory result must have a valid format
- ❖ Wearable readings must be within reasonable limits
- ❖ Mandatory fields cannot be NULL

Possible Data Quality Issues

Issue	Solution
Missing patient information	Data validation
Duplicate patient records	Deduplication
Incorrect date formats	Data standardization
Invalid laboratory results	Validation rules
Inconsistent patient information	Data cleansing
Missing wearable readings	Data quality checks

---

9. Analytics and AI Applications

Analytics Dashboards

1. Patient Waiting Time Dashboard

Helps management identify departments with long waiting times.

2. Patient Flow Dashboard

Shows:

- Number of patients
- Admissions

- Discharges
- Emergency visits
- Department-wise patient volume

3. Doctor Performance Dashboard

Shows:

- Number of consultations
- Average consultation time
- Patient load
- Appointment completion rate

4. Laboratory Dashboard

Shows:

- Number of tests
- Abnormal results
- Department-wise tests
- Test processing time

5. Patient Risk Dashboard

Shows:

- High-risk patients
- Risk score
- Important health indicators
- Recent abnormal readings

AI Applications

1. High-Risk Patient Prediction

AI can analyze:

- Patient history
- Laboratory results
- Vital signs
- Wearable data
- Doctor observations

It can identify patients who may require closer monitoring.

2. Waiting Time Prediction

AI can predict expected waiting times based on:

- Number of appointments
- Doctor availability
- Department workload
- Historical waiting times

This can help the hospital reduce patient waiting time.

3. Patient Readmission Prediction

The model can identify patients who may have a higher probability of returning to the hospital after discharge.

4. Early Warning System

Wearable-device data can be analyzed to identify unusual changes in vital signs and alert healthcare staff for further assessment.

5. Patient Segmentation

Patients can be grouped based on:

- Age
- Medical history
- Treatment requirements
- Risk level
- Healthcare utilization

Conclusion

The hospital can use this data engineering pipeline to bring data from registration, appointments, laboratory systems, wearable devices, and consultation notes into one integrated platform. The data is ingested using batch, stream, and file-based methods, cleaned through ETL, stored in a Data Lake and Data Warehouse, and then used for dashboards and AI applications. This helps hospital management make data-driven decisions, improve operational efficiency, reduce waiting times, and identify patients who may need closer attention.